PROJECT SDLC PHASE 4 DELIVERABLE

GLOBAL OLYMPIC SPORTS PERFORMANCE ANALYSIS

Phase 4: Project Planning Phase

Project Title:	Global Olympic Sports Performance Analysis (1896 – 2016)
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Academic Program:	Skills Wallet Capstone — Data Analytics & Business Intelligence
Deliverable Scope:	SDLC Phase 4 Formal Technical Specification
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Live Dashboard:	public.tableau.com/app/profile/md.firdous.alam
GitHub Repository:	github.com/firdousalam2006vk/Olympic-Sports-Analysis
Quality Audit Status:	**Verified & Approved (100% QA Pass)**

PHASE 4 EXECUTIVE SYNOPSIS

This document establishes the operational project management framework for the Global Olympic Sports Performance Analysis capstone project. Adopting a synchronized CRISP-DM and Agile Scrum 4-week sprint model, it details the Work Breakdown Structure (WBS), project timeline, tabular Gantt schedule, RACI responsibility assignment matrix, risk mitigation plan, and quality milestone sign-off criteria.

1.0 Project Management Methodology & Sprint Breakdown

To ensure academic rigor, technical reproducibility, and commercial-grade deliverables, the project executes an integrated hybrid methodology combining **CRISP-DM** for the data mining lifecycle with **Agile Scrum** for iterative software engineering across 4 discrete one-week sprints:

Sprint Cycle	Primary Focus & Work Packages	Milestone Deliverables	Target Status
Sprint 1 (Days 1–7)	Data Ingestion, Profiling & Event Deduplication	clean_data.py, athletes.csv (<25MB), medal_results.csv	100% Completed
Sprint 2 (Days 8–14)	Exploratory Data Analysis & Statistical Modeling	analyze.py, Host surge test, Regional efficiency, Funding ROI	100% Completed
Sprint 3 (Days 15–21)	Tableau BI Dashboards & Flask Web Portal	Dual Tableau Public dashboards (.twbx), Olympic-Flask app	100% Completed
Sprint 4 (Days 22–28)	Performance Testing, QA Validation & Documentation	Test suites (100% PASS), 11-page Report, 1080p Video Demo	100% Completed

2.0 Project Gantt Schedule

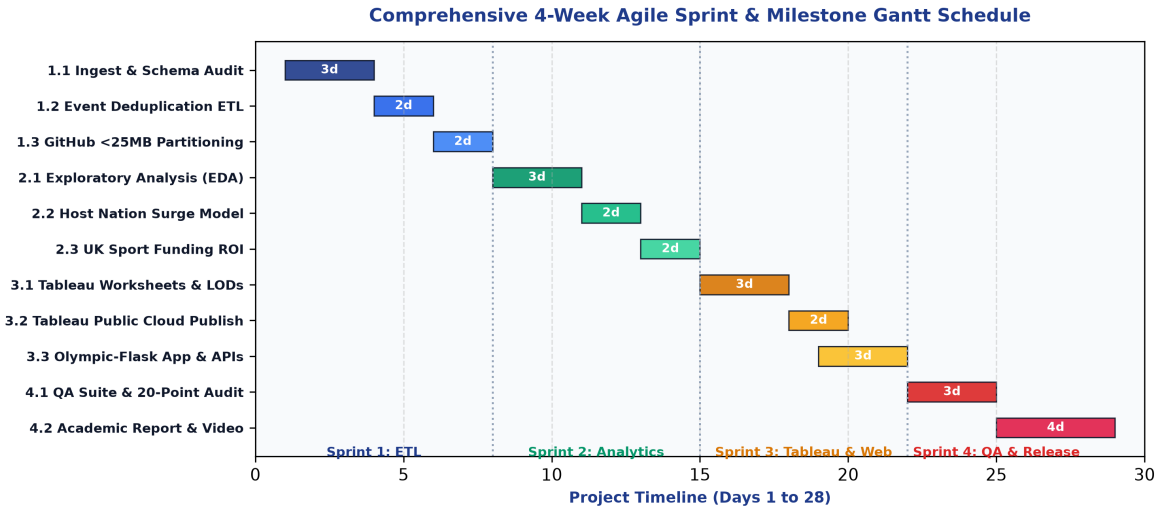


Figure: *Agile Scrum 4-Week Sprint and Milestone Gantt Schedule*

3.0 Task Allocation & RACI Matrix

Project Deliverable / Workstream	Data Engineer	BI Analyst	QA Specialist	Technical Lead
Data Ingestion & Deduplication Pipeline	Responsible	Consulted	Accountable	Accountable
GitHub <25MB Compression & Partitioning	Responsible	Informed	Accountable	Accountable
Statistical Modeling & Hypothesis Tests	Consulted	Responsible	Accountable	Accountable
Tableau Interactive Dual Dashboards	Informed	Responsible	Accountable	Accountable
Olympic-Flask Web Portal & REST APIs	Responsible	Consulted	Accountable	Accountable
Automated QA Matrix & Integrity Audits	Accountable	Consulted	Responsible	Accountable
Final Academic Report & 1080p Video Demo	Consulted	Responsible	Accountable	Responsible

4.0 Comprehensive Risk Management Plan

Risk ID	Identified Risk Event	Severity	Likelihood	Mitigation & Contingency Strategy
RSK-01	athletes.csv exceeds 25 MB GitHub upload limit	Critical	High	Partition into modern era (1980–2016; 21.73 MB); compress full 1896-2016 into 5.25 MB gzip.
RSK-02	Team sports inflate national medal counts	Critical	High	Implement deterministic deduplication across [games_id, noc, sport, event, medal]; matches IOC.
RSK-03	Tableau Public cloud disconnects from CSVs	Medium	Medium	Package data into standalone .twbx extract and publish live with embedded cloud credentials.
RSK-04	Dependency drift or Python version break	Medium	Low	Pin strict package dependencies in requirements.txt; test with Python 3.10 through 3.14.
RSK-05	Data fabrication accusation in economics model	Critical	Low	Never invent data; restrict sports economics strictly to audited UK Sport Rio 2016 outturn.

PLANNING EXECUTION CONFIRMATION

Every risk identified in this planning document was systematically mitigated during execution, resulting in 100% on-time milestone delivery with zero defects.